

# Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

For the MCQ-label correctness narrowly, MILU has reasonable provenance: answers are inherited from online exam portals where 'subject experts ensure the accuracy of the answers' [\[Q26\]](#), with AI4Bharat volunteer manual audits [\[Q87\]](#) and researcher review of subject-tag clusters [\[Q41\]](#). However, the paper reports no annotator demographics, no inter-annotator agreement statistics, and no formal QA protocol beyond manual sampling. Critically, the deployment's core output (Hindi rationale) has no ground-truth labels in MILU at all. Dataset analysis surfaces additional concerns: cross-language item duplication [\[D14, D15, D29\]](#) and malformed items with truncated stems [\[DATASET-D34, D35\]](#) suggest QA imperfections at the item level. ~55% of subject labels were assigned via machine translation + GPT-4O-MINI [\[Q37, Q38\]](#), an automated process that may not preserve Hindi-specific terminology fidelity.

| ID                  | Evidence                                                                                                                                                                                                                                                               |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q26</a> | "These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)                                                                                                                  |
| <a href="#">Q87</a> | "We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)                                                                                                              |
| <a href="#">Q41</a> | "We manually review these clusters and assign appropriate subject labels." (p.4)                                                                                                                                                                                       |
| <a href="#">Q37</a> | "Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)                                                                                               |
| <a href="#">Q38</a> | "To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)                                                     |
| <a href="#">D34</a> | The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed                                                                                                                             |
| <a href="#">D14</a> | What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline |
| <a href="#">D15</a> | 1991 में वित्तीय संकट को दूरगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline   |
| <a href="#">D29</a> | ഭരണീയ അടിയന്തരാവസ്ഥ പര്യന്താപിക്കാൻ 'സാധു കലാപം' എന്ന് പദം ഭരണഘടനയിൽ ചേർത്തത് എപ്പോൾ? ... 44-ാമത് ഭരണഘടനാ ഭേദഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication                           |
| <a href="#">D35</a> | ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിൽപ്പം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages                                                                                                                   |

## Strengths

- Answer keys derive from established Indian competitive-exam ecosystems where subject experts validate correctness at source [\[Q26\]](#).
- Multi-stage manual review applied to subject-tag clusters [\[Q41\]](#) and a manual audit pass [\[Q87\]](#).
- Constitutional, historical, and GK answer labels in the observed Hindi sample appear factually correct [\[DATASET-D1, D2, D22\]](#).

| ID | Evidence |
|----|----------|
|----|----------|

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|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q26</b> | "These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)                                                                       |
| <b>Q41</b> | "We manually review these clusters and assign appropriate subject labels." (p.4)                                                                                                                                            |
| <b>Q87</b> | "We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)                                                                   |
| <b>D1</b>  | राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic |
| <b>D2</b>  | मुगल ग्रंथ 'मासिर-ए-आलमगीरी' के रचयिता कौन हैं? ... साकी मुस्तईद खान — Mughal-period history question in Hindi — directly relevant to North India medieval history                                                          |
| <b>D22</b> | What was the first capital of the Bahamani Kingdom? ... Gulbarga — India-specific historical knowledge (Deccan Sultanate) — same question appears across all languages                                                      |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** MCQ answer labels reflect Indian exam-portal subject-expert consensus [Q26], which is reasonable for central-exam content. However, regional stakeholder perspectives on rationale quality have no representation since rationale labels do not exist.

**OC-2:** INSUFFICIENT DOCUMENTATION on annotator-population disagreement risk for MCQ labels; annotator demographics are not reported beyond 'AI4Bharat volunteers' [Q87].

**OC-3:** Paper does not provide a Datasheet or Data Statement with annotator demographics — gap for assessing convergent validity.

**OC-4:** Re-annotation by Hindi-medium subject experts would be needed to (a) validate translation quality on the ~25%+ translated Hindi subset, (b) flag items where the answer is region-sensitive, (c) generate ground-truth Hindi rationales.

**OC-5:** INSUFFICIENT DOCUMENTATION on aggregation methods; the paper describes manual cluster review [Q41] but not how disagreements were resolved.

**OC-6:** Label issues: (a) no IAA statistics; (b) ~55% of topic labels assigned by automated MT+GPT pipeline [Q37, Q38]; (c) cross-language duplicates introduce identical labels across translated copies [D14, D15]; (d) malformed items have labels whose validity is questionable [DATASET-D34, D35]; (e) rationale ground-truth wholly absent.

| ID         | Evidence                                                                                                                                                                                                           |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q26</b> | "These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)                                                              |
| <b>Q87</b> | "We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)                                                          |
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| <b>Q41</b> | "We manually review these clusters and assign appropriate subject labels." (p.4)                                                                                                                                   |

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| Q37 | "Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)                                                                                               |
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### Information Gaps

- Annotator demographics, IAA statistics, and disagreement-resolution protocols are not reported.
- Translation-quality verification for the Hindi partition is not documented.
- No Hindi rationale ground truth exists in the benchmark.

### Requires Expert Verification

- A Hindi-medium subject-matter expert should audit a representative Hindi sample for label correctness, especially for translated items and items with stems that may have been corrupted by RC-passage stripping.

### Remediation

**Gap 1: No annotator demographics, IAA statistics, or formal QA protocol reported; ~55% of topic labels assigned by automated GPT-4O-MINI [Q37, Q38]; malformed items observed [DATASET-D34, D35].**

**Recommendation:** Conduct a Hindi-medium expert audit on a stratified sample of the Hindi test split: verify answer correctness, translation-register quality, item completeness, and subject-tag accuracy. Report IAA. Filter or repair malformed items prior to deployment-grade evaluation.

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|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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